TREEWALK NYC • STEAM LESSON PLAN

# Norway Maple

*Acer platanoides*

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**Species background:** Native to continental Europe and western Asia; brought to North America in the 1750s by botanist John Bartram and later planted extensively as a street tree. Snap a leaf stem and look for a drop of milky white sap -- this is one of the few reliable ways to distinguish Norway maple from native maples at a glance, alongside its wide, dense crown.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

# Norway Maple: Identify Norway maple by its hand-shaped leaf and find its milky sap clue.

**Standards connection:** NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

**Materials:** Coloring sheet, crayons.

## Teacher background

Check if the leaf looks like a hand with pointy fingers -- Norway maple has five points, like five fingers spread wide.

## Procedure

1. Engage: Trace your own hand, then compare it to the leaf shape.
2. Explore: Count the leaf's points together like counting fingers.
3. Explain: Explain that a grown-up can gently snap a leaf stem to look for a drop of milky sap, one clue that helps tell this maple apart from others.
4. Art: Trace around a fallen leaf, then color each of its five points a different color like a pinwheel.
5. Evaluate: Student can count and point out five leaf points and describe the leaf as hand-shaped.

## Discussion questions

What other things in nature have five parts, like this leaf?

## Assessment

Student correctly counts five points and matches them to their own hand.

## Extension

Compare this leaf to another maple leaf found on the walk -- same number of points?

GRADE BAND 3-5

# Norway Maple: Use the opposite-branching pattern and milky sap to distinguish Norway maple from other trees.

**Standards connection:** NGSS 3-LS4-3; National Core Arts Standards VA:Cr2 (3-5)

**Materials:** Coloring sheet, colored pencils.

## Teacher background

Most trees branch in a staggered zigzag, but maples, ashes, and a few others branch in exact opposite pairs -- a real shortcut arborists use in the field (mnemonic: MAD Horse -- Maple, Ash, Dogwood, Horsechestnut).

## Procedure

1. Engage: Ask students to look at how branches attach to a twig -- staggered or paired?
2. Explore: Check a real or pictured Norway maple twig for opposite branching.
3. Explain: Teach the MAD Horse mnemonic and explain what makes opposite branching useful for ID.
4. Art: Color the leaf outline, then create a leaf-shape stencil for a repeated print.
5. Evaluate: Student names one other MAD Horse tree and explains what opposite branching means.

## Discussion questions

Why might arborists want a fast field-ID shortcut instead of examining the whole tree?

## Assessment

Correctly explains opposite branching and names at least one other MAD Horse species.

## Extension

Find one opposite-branching and one alternate-branching tree on your walk and sketch both twig patterns.

GRADE BAND 6-8

# Norway Maple: Explain opposite branching as a field-ID trait and evaluate Norway maple's invasive impact.

**Standards connection:** NGSS MS-LS2-4; National Core Arts Standards VA:Cr2/Re7 (6-8)

**Materials:** Coloring sheet, print-making materials (stencil, ink or paint).

## Teacher background

Norway maple is now considered invasive in many parts of the northeastern US and Canada because its dense shade and heavy seed production can crowd out native forest seedlings. Its opposite branching pattern is shared with ash, dogwood, and horsechestnut.

## Procedure

1. Engage: What does 'invasive' mean, and how is it different from just 'introduced'?
2. Explore: Research how dense shade and seed production give Norway maple a competitive edge over native seedlings.
3. Explain: Connect the opposite-branching ID trait to the ecological impact discussion.
4. Art: Create a leaf-shape stencil print pattern, alternating colors to echo the plant's opposite-branching rhythm.
5. Evaluate: Write a paragraph explaining why a popular street tree can also be considered invasive in nearby forests.

## Discussion questions

Should cities stop planting a species once it's known to be invasive nearby, even if it's a good street tree? What tradeoffs are involved?

## Assessment

Paragraph correctly explains the invasive mechanism (shade + seed production) with specific detail.

## Extension

Research one native maple that could serve a similar street-tree role without the invasive risk.

GRADE BAND 9-12

# Norway Maple: Design a field study comparing Norway maple's phenology to a native tree's, and evaluate its competitive advantage.

**Standards connection:** NGSS HS-LS2-6; National Core Arts Standards Proficient-level Creating/Responding (9-12)

**Materials:** Coloring sheet, notebook for weekly observation log, fine-liner pens.

## Teacher background

Norway maple leafs out earlier and holds leaves later than many native trees, extending its photosynthetic season -- a phenological advantage that helps explain its invasive spread into native forests.

## Procedure

1. Engage: How could 'leafing out a week earlier' give a tree species a measurable competitive advantage?
2. Explore: Design a simple field study comparing leaf-out dates between Norway maple and a native oak on the same block over several weeks.
3. Explain: Discuss extended photosynthetic season as a mechanism for outcompeting native seedlings for light.
4. Art: Do a comparative botanical sketch of Norway maple next to a native maple leaf, annotating differences in lobe shape and sinus depth with precise line work.
5. Evaluate: Submit the field-study design (or collected data if time allows), the comparative sketch, and a short written conclusion.

## Discussion questions

What experimental controls would make your leaf-out comparison more scientifically rigorous?

## Assessment

Rubric covering study design validity, sketch precision, and reasoning in the written conclusion.

## Extension

Research current NYC or regional policy on Norway maple planting and evaluate whether it reflects this phenological evidence.

## SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.